

WIND TUNNEL TESTING PROCEDURE

1. Competition evaluation metric

The evaluation metric is based on the energy criterion proposed by Prof. Ryszard Szwaba. It is an idealised annual energy production which takes into account wind data of the southern Baltic Sea [1]. It is calculated according to the following formula:

$$E = T \int_{V_{cut-in}}^{V_{cut-out}} P(V) f(V) dV ,$$

where T is the number of hours in a year, $P(V)$ is the power curve of the wind turbine, $f(V)$ is the Weibull cumulative distribution function representing the 50-year-averaged wind speed of the southern Baltic Sea [1], V is the wind speed at the steady-state air flow, $V_{cut-in} = 3$ m/s, and $V_{cut-out} = 12$ m/s.

The Weibull cumulative distribution function is given by:

$$f(V) = 1 - \exp \left[- \left(\frac{V}{c} \right)^k \right] ,$$

where monthly shape parameters k and scale parameters c are given in **Table 1**.

Table 1. Monthly long-term shape parameters k and scale parameters c [1].

Month	Shape k [m/s]	Scale c [m/s]
January	2.44	9.40
February	2.41	8.97
March	2.29	8.52
April	2.24	7.16
May	2.22	6.70
June	2.17	6.66
July	2.12	6.84
August	2.20	6.87

September	2.27	7.61
October	2.40	8.89
November	2.51	9.29
December	2.52	9.37

1.1. First stage of the wind tunnel testing procedure

In the first stage of the wind tunnel testing procedure, a power curve of the wind turbine model $P(V)$ will be evaluated. For the steady wind velocity conditions, starting from 3 m/s, the power will be determined at a given rotation speed controlled by the generator. The maximum generated power will be selected for further evaluation. Next, wind velocity will be increased by 1 m/s, and the same procedure will be applied. The procedure will be repeated until the wind velocity reaches 12 m/s.

Important remark #1:

The wind turbine rotor will be evaluated in the range of rotation speeds up to rpm_{max} which should be provided by the designers (teams)! Please consider safe operation of your design at high rotation speeds (e.g. 2000 rpm) and be reasonable about rpm_{max} .

Important remark #2:

If your turbine rotor is not self-starting at low wind velocity conditions, the generator will help with initial acceleration. Due to negligible energy needed for the initial acceleration of the turbine rotor, we have decided that this energy will not be deducted from the annual energy production.

1.2. Second stage of the wind tunnel testing procedure

The second stage of the wind tunnel testing procedure is the processing of the acquired data. In order to ensure full transparency of the competition, the script in Python ([turbine_production.py](#)) and its equivalent in MATLAB ([turbine_production.m](#)) have been prepared by Dr Janusz Telega. These scripts are implementations of the evaluation metric based on the annual energy production criterion (note that, for simplicity, the linear interpolation is used). Based on this single metric, the winners of the competition will be announced.

Bibliography

- [1] W. Zhang, J. Harff, R. Schneider, Analysis of 50-year wind data of the southern Baltic Sea for modeling coastal morphological evolution - A case study from the Darss-Zingst Peninsula, Oceanologia. 53 (2011) 489–518. doi:10.5697/oc.53-1-TI.489.